

Friday, Apr 26

Nonlinear Regression With Random Effects

Example: The model we specified for the `Sitka` data can be written as

$$E(Y_{ij}) = \beta_0 + \beta_1 o_{ij} + \beta_2 w_{ij} + \beta_3 o_{ij} w_{ij} + \delta_i + \gamma_i w_{ij},$$

where o_{ij} is an indicator for if the observation is from the ozone treatment condition and w_{ij} is weeks. We can also write this model as

$$E(Y_{ij}) = \underbrace{\beta_0 + \beta_1 o_{ij} + \delta_i}_{\beta_{0ij}} + \underbrace{(\beta_2 + \beta_3 o_{ij} + \gamma_i)}_{\beta_{1ij}} w_{ij},$$

or

$$E(Y_{ij}) = \beta_{0ij} + \beta_{1ij} w_{ij},$$

to show that the model assumes a linear relationship between expected size and weeks, but where the “intercept” β_{0ij} depends on the treatment condition and tree, and the “slope” β_{1ij} depends on the treatment condition and tree because

$$\beta_{0ij} = \beta_0 + \beta_1 o_{ij} + \delta_i$$

$$\beta_{1ij} = \beta_2 + \beta_3 o_{ij} + \gamma_i.$$

Models with random effects written in this way are sometimes called “random coefficient” models. The coefficients β_{0ij} and β_{1ij} are random (due to δ_i and γ_i) but may also depend on one or more explanatory variables (such as treatment condition via o_{ij}).

The `nlme` function from the **nlme** package can estimate a linear or nonlinear regression model with random coefficients (assuming a normally-distributed response variable and random parameters). We estimated a model for the `Sitka` data as follows.

```
library(MASS)
library(lme4)
m <- lmer(exp(size) ~ treat * I(Time/7) + (1 + I(Time/7) | tree),
  data = Sitka, REML = FALSE)
```

Warning in checkConv(attr(opt, "derivs"), opt\$par, ctrl = control\$checkConv, : Model failed to converge with max|grad| = 0.0042734 (tol = 0.002, component 1)

```
summary(m)$coefficients
```

	Estimate	Std. Error	t value
(Intercept)	-305.123	31.249	-9.764
treatozone	110.675	37.796	2.928
I(Time/7)	17.565	1.685	10.425
treatozone:I(Time/7)	-5.516	2.038	-2.707

I am using `REML = FALSE` to use maximum likelihood rather than *restricted* maximum likelihood (REML) for estimation so that we can compare the results with `nlme`, which only uses maximum likelihood. We are going to assume we can ignore the warning.

```
library(nlme)
m <- nlme(exp(size) ~ b0 + b1 * I(Time/7),
  fixed = b0 + b1 ~ treat,
  random = b0 + b1 ~ 1 | tree,
  start = c(0,0,0,0), data = Sitka)
summary(m)
```

Nonlinear mixed-effects model fit by maximum likelihood

Model: $\exp(\text{size}) \sim b_0 + b_1 * I(\text{Time}/7)$

Data: Sitka

AIC BIC logLik

3947 3978 -1965

Random effects:

Formula: $\text{list}(b_0 \sim 1, b_1 \sim 1)$

Level: tree

Structure: General positive-definite, Log-Cholesky parametrization

	StdDev	Corr
b0.(Intercept)	148.699	b0.(I)
b1.(Intercept)	8.266	-0.987
Residual	19.575	

Fixed effects: $b_0 + b_1 \sim \text{treat}$

	Value	Std.Error	DF	t-value	p-value
b0.(Intercept)	-305.12	31.41	313	-9.714	0.0000
b0.treatozone	110.68	37.99	313	2.913	0.0038
b1.(Intercept)	17.56	1.69	313	10.371	0.0000
b1.treatozone	-5.52	2.05	313	-2.693	0.0075

Correlation:

	b0.(I)	b0.trt	b1.(I)
b0.treatozone	-0.827		
b1.(Intercept)	-0.980	0.810	
b1.treatozone	0.810	-0.980	-0.827

Standardized Within-Group Residuals:

	Min	Q1	Med	Q3	Max
	-2.96548	-0.38922	-0.05501	0.38267	4.82179

Number of Observations: 395

Number of Groups: 79

The `nlme` function is like `nls` in that it needs starting values for the (fixed) parameters, but since the model is linear we do not need particularly good starting values.

Example: Now consider a *nonlinear* regression model with random effects for the `Loblolly` data that come with R.

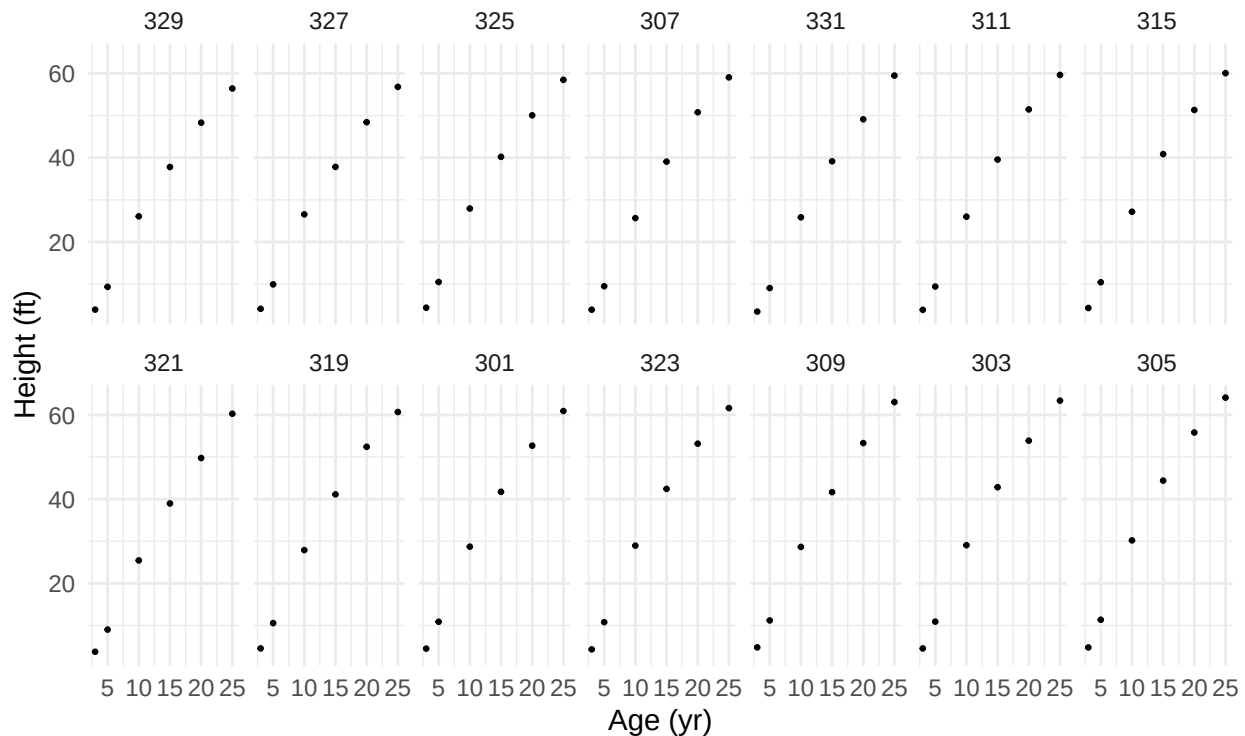
```
head(Loblolly)
```

Grouped Data: $\text{height} \sim \text{age} \mid \text{Seed}$

	height	age	Seed
1	4.51	3	301
15	10.89	5	301
29	28.72	10	301
43	41.74	15	301

```
57 52.70 20 301
71 60.92 25 301
```

```
p <- ggplot(Loblolly, aes(x = age, y = height)) +
  geom_point(size = 0.5) + facet_wrap(~ Seed, ncol = 7) +
  ylab("Height (ft)") + xlab("Age (yr)") + theme_minimal()
plot(p)
```



Suppose we want to estimate the nonlinear growth model

$$E(H) = \theta_1 + (\theta_2 - \theta_1)e^{-a \log(2)/\theta_3},$$

where H and a are height and age, respectively, θ_1 is the asymptote as $a \rightarrow \infty$, and θ_2 is an “intercept” parameter, and θ_3 is the age at which the tree is half way between $E(H) = \theta_2$ and $E(H) = \theta_1$. To allow for differences between trees with respect to θ_1 and θ_3 (but not θ_2) we could write the model as

$$E(H_{ij}) = \theta_{1i} + (\theta_2 - \theta_{1i})e^{-a_{ij} \log(2)/\theta_{3i}},$$

where H_{ij} and a_{ij} are now the height and age of the j -th observation of the i -th tree.

```
m <- nlme(height ~ t1 + (t2 - t1) * exp(-age * log(2)/t3),
  fixed = t1 + t2 + t3 ~ 1,
  random = t1 + t3 ~ 1 | Seed,
  start = c(t1 = 100, t2 = 0, t3 = 15),
  data = Loblolly)
summary(m)
```

```
Nonlinear mixed-effects model fit by maximum likelihood
Model: height ~ t1 + (t2 - t1) * exp(-age * log(2)/t3)
Data: Loblolly
   AIC   BIC logLik
239.1 256.1 -112.5
```

```

Random effects:
Formula: list(t1 ~ 1, t3 ~ 1)
Level: Seed
Structure: General positive-definite, Log-Cholesky parametrization
      StdDev Corr
t1      5.4054 t1
t3      1.2334 0.769
Residual 0.6462

```

```

Fixed effects:  t1 + t2 + t3 ~ 1
      Value Std.Error DF t-value p-value
t1 101.03    2.4708 68   40.89      0
t2  -8.71    0.2850 68  -30.55      0
t3  17.48    0.6289 68   27.79      0
Correlation:
  t1    t2
t2 0.615
t3 0.918 0.698

```

```

Standardized Within-Group Residuals:
      Min      Q1      Med      Q3      Max
-1.88049 -0.59420 -0.04009  0.70784  1.47988

```

```

Number of Observations: 84
Number of Groups: 14

```

We can plot the estimated growth curves (both per tree and average) as follows.

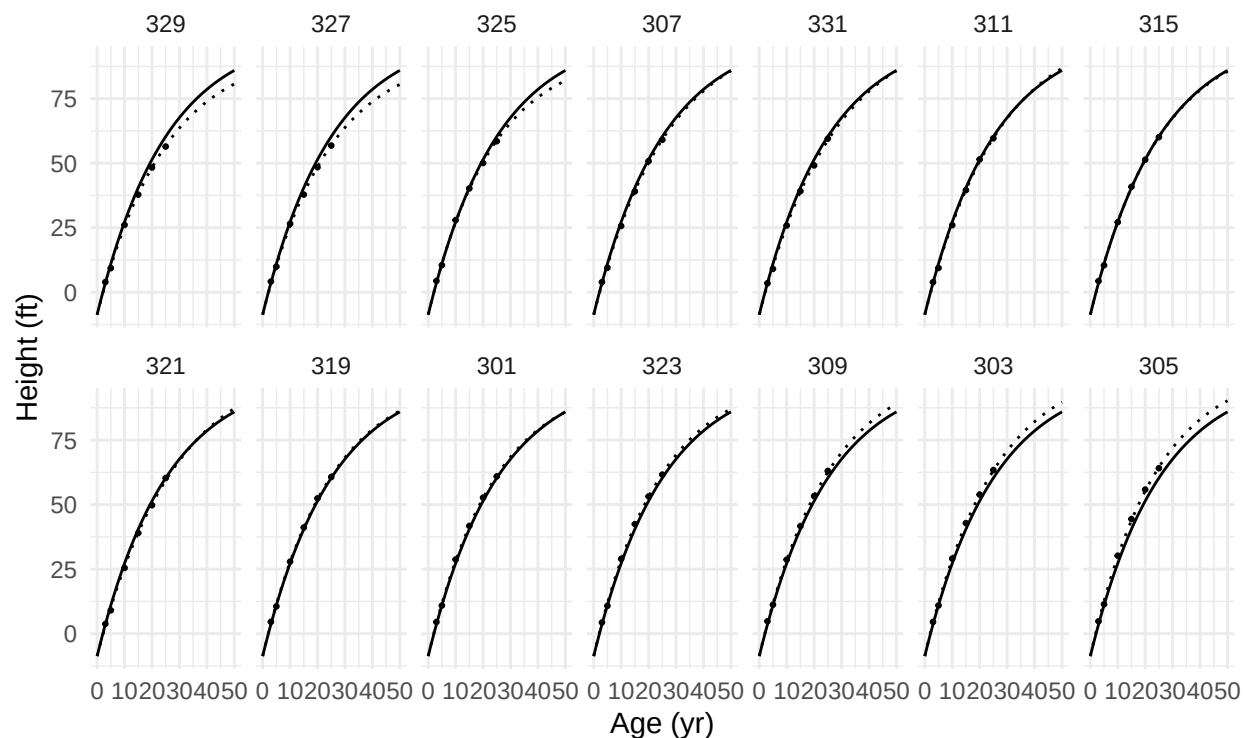
```

d <- expand.grid(age = seq(0, 50, length = 100), Seed = unique(Loblolly$Seed))

d$yhat.ind <- predict(m, newdata = d, level = 1) # individual tree
d$yhat.avg <- predict(m, newdata = d, level = 0) # average tree

p <- ggplot(Loblolly, aes(x = age, y = height)) +
  geom_line(aes(y = yhat.ind), data = d, linetype = 3) +
  geom_line(aes(y = yhat.avg), data = d) +
  geom_point(size = 0.5) + facet_wrap(~ Seed, ncol = 7) +
  ylab("Height (ft)") + xlab("Age (yr)") + theme_minimal()
plot(p)

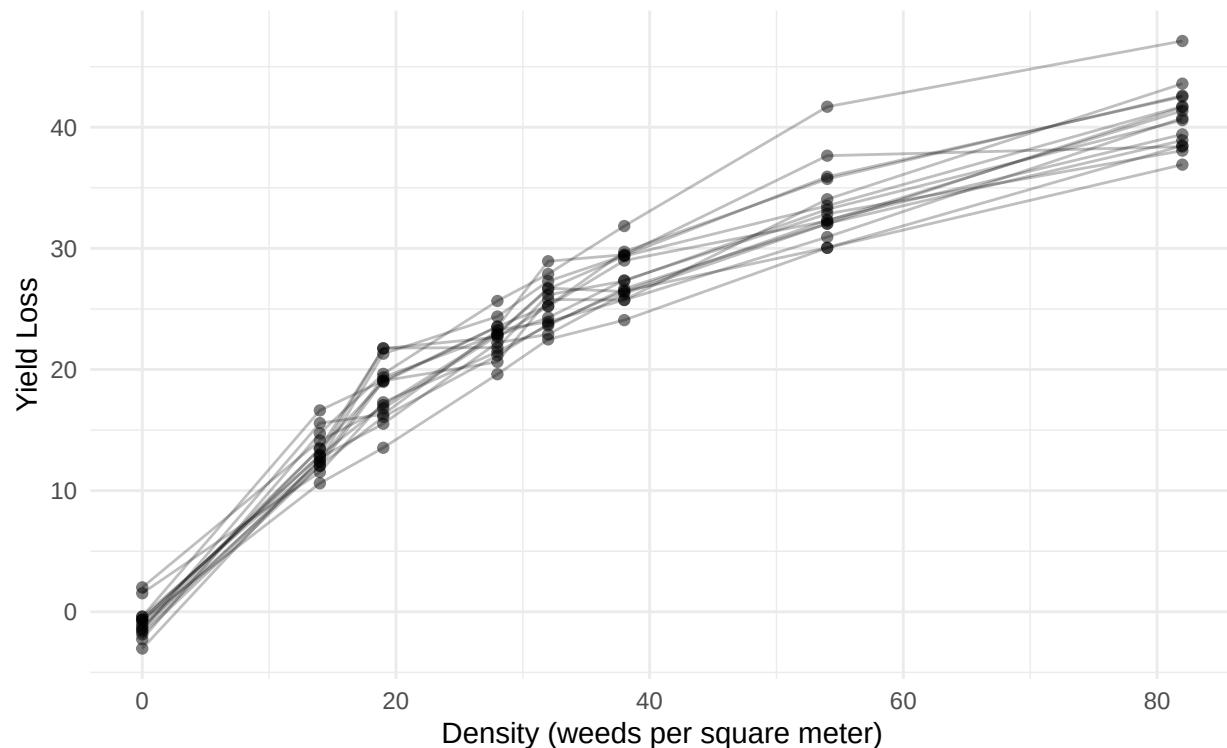
```



Example: Here are some data from an experiment using a randomized block design on the effect of weed density on yield loss of sunflowers.

```
yieldloss <- read.csv("https://raw.githubusercontent.com/OnofriAndreaPG/agroBioData/master/YieldLossB.csv")
```

```
p <- ggplot(yieldloss, aes(x = density, y = yieldLoss)) + theme_minimal() +
  labs(x = "Density (weeds per square meter)", y = "Yield Loss") +
  geom_line(aes(group = block), alpha = 0.25) + geom_point(alpha = 0.5)
plot(p)
```



The model suggested for these data has the same form as the Michaelis-Menten model, but with random effects to account for the effect of block.

```
m <- nlme(yieldLoss ~ alpha * density / (beta + density),
  fixed = list(alpha ~ 1, beta ~ 1),
  random = alpha + beta ~ 1 | block,
  start = c(alpha = 60, beta = 30), data = yieldloss)
```

```
summary(m)$tTable
```

	Value	Std.Error	DF	t-value	p-value
alpha	67.64	1.826	104	37.05	9.864e-62
beta	54.48	2.524	104	21.58	3.335e-40

```
intervals(m)
```

Approximate 95% confidence intervals

Fixed effects:

	lower	est.	upper
alpha	64.05	67.64	71.23
beta	49.51	54.48	59.44

Random Effects:

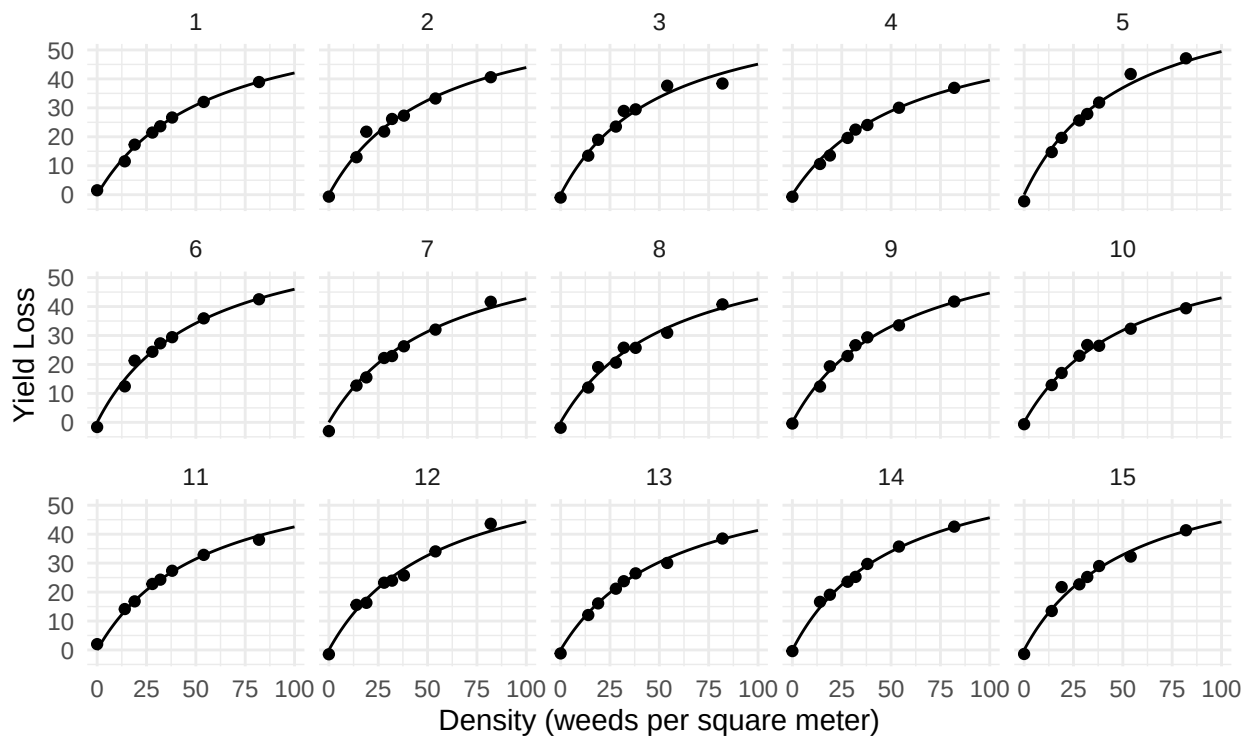
	Level: block	lower	est.	upper
sd(alpha)		1.4639	3.34722	7.6534
sd(beta)		0.8946	3.61161	14.5804
cor(alpha,beta)		-0.6292	0.01093	0.6422

Within-group standard error:

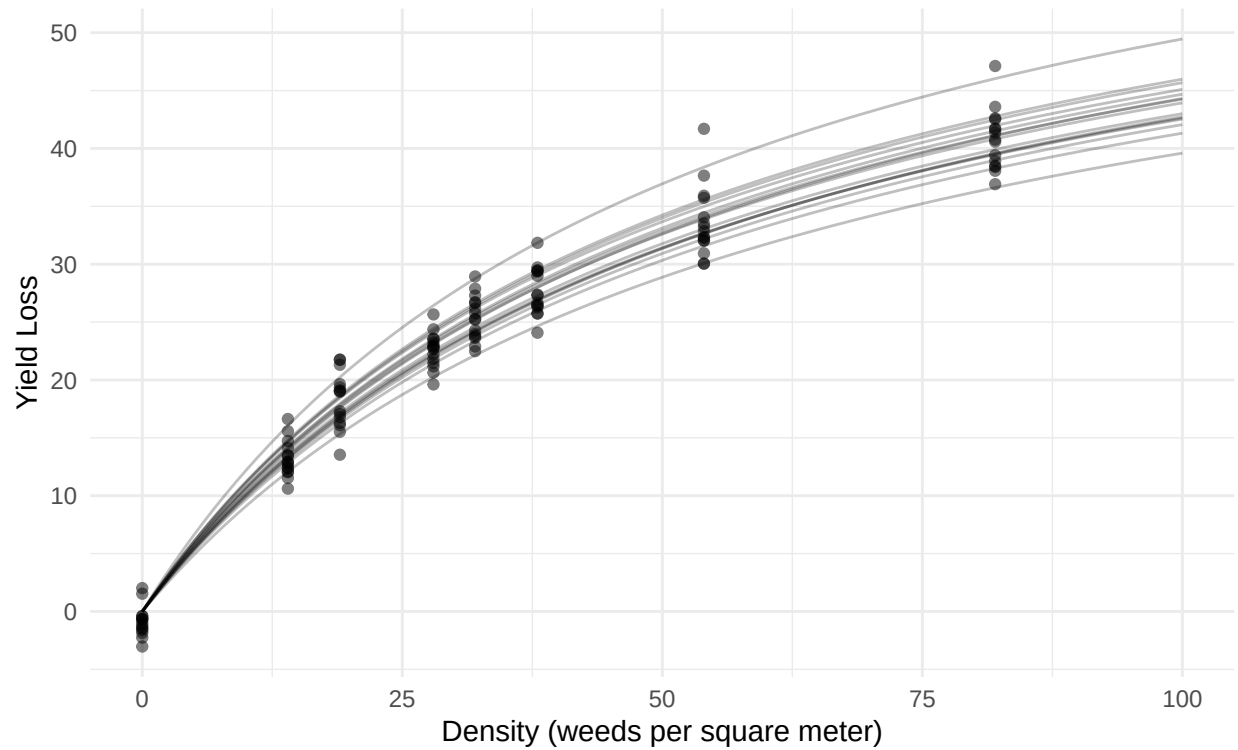
```
lower est. upper
1.246 1.432 1.644
```

```
d <- expand.grid(density = seq(0, 100, length = 100), block = 1:15)
d$yhat <- predict(m, newdata = d)
```

```
p <- ggplot(yieldloss, aes(x = density, y = yieldLoss)) + theme_minimal() +
  labs(x = "Density (weeds per square meter)", y = "Yield Loss") +
  geom_point() + geom_line(aes(y = yhat), data = d) +
  facet_wrap(~ block, ncol = 5)
plot(p)
```



```
p <- ggplot(yieldloss, aes(x = density, y = yieldLoss)) + theme_minimal() +
  labs(x = "Density (weeds per square meter)", y = "Yield Loss") +
  geom_point(alpha = 0.5) + geom_line(aes(y = yhat, group = block), alpha = 0.25, data = d)
plot(p)
```



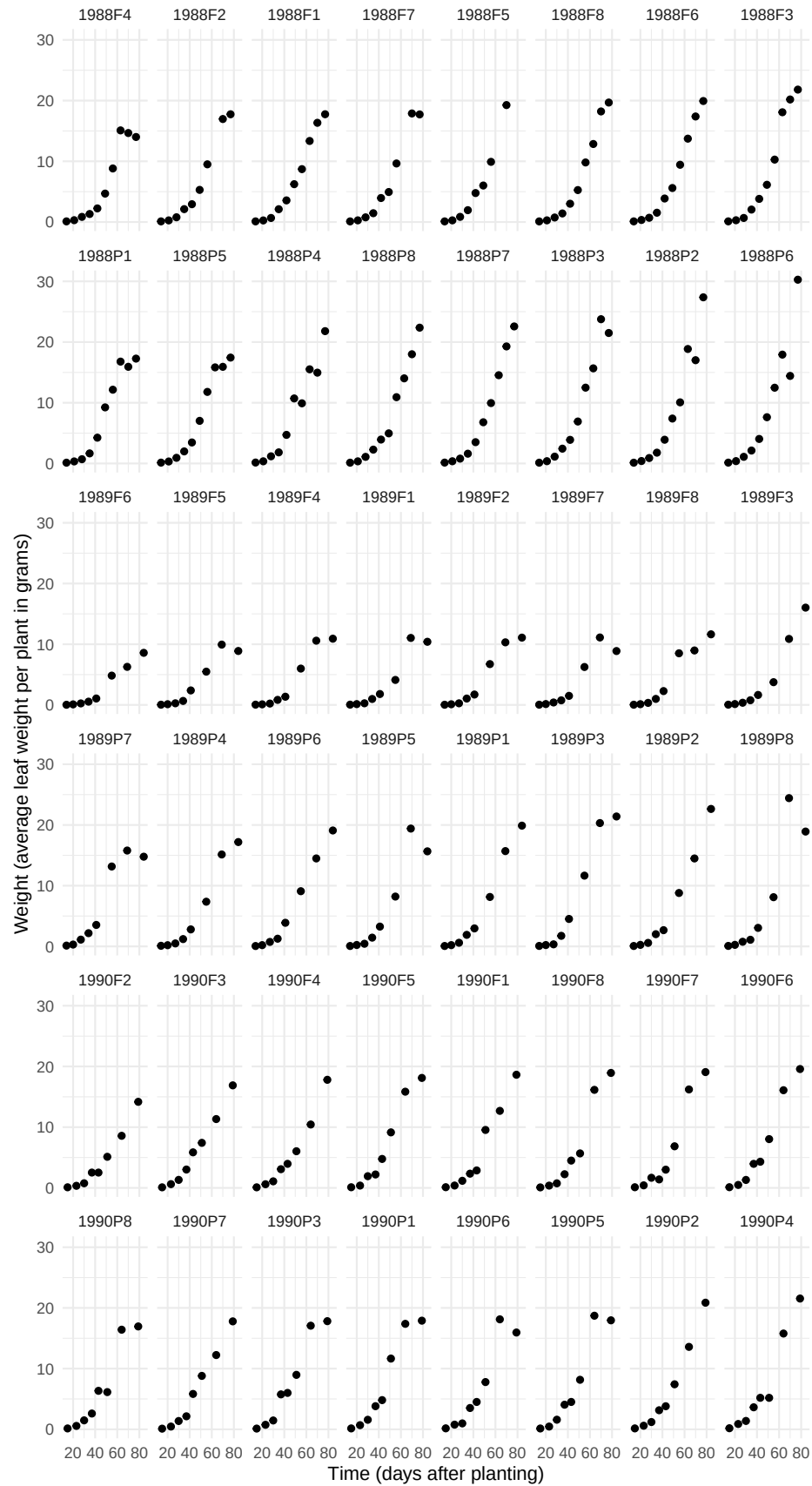
Example: The data frame `Soybean` from the `nlme` package has data from an experiment looking at soybean growth.

```
head(Soybean)
```

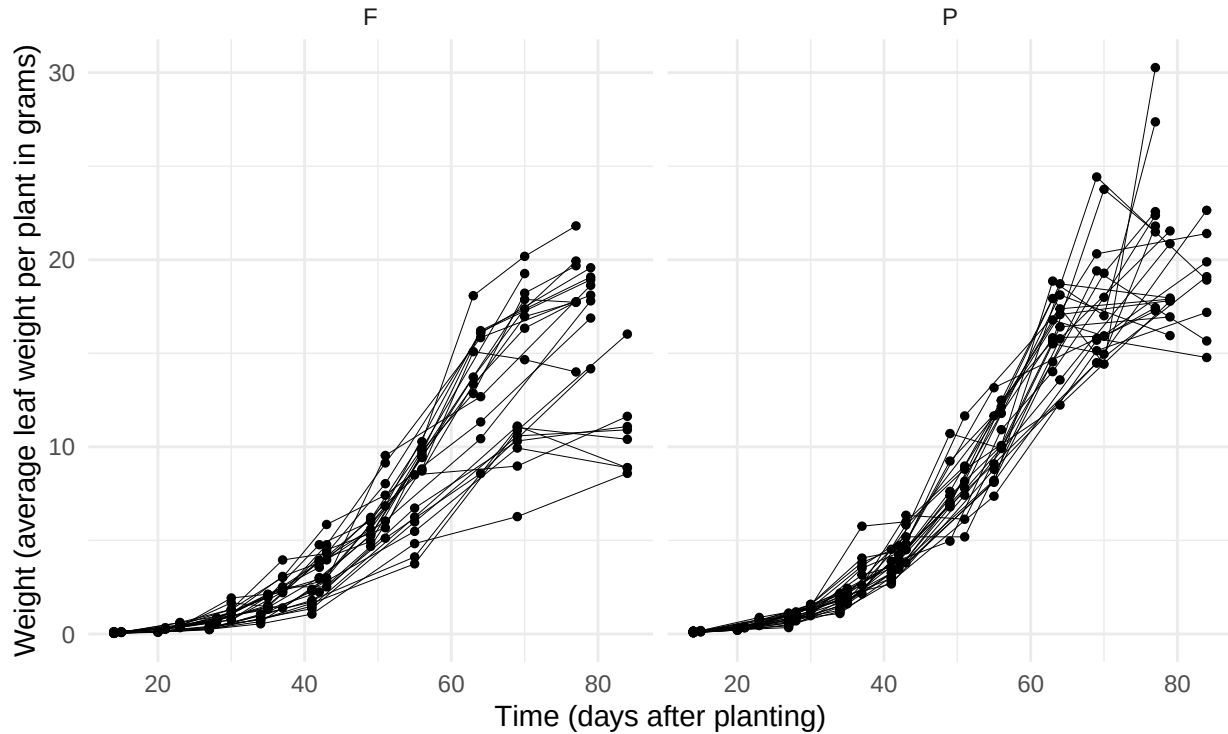
```
Grouped Data: weight ~ Time | Plot
```

	Plot	Variety	Year	Time	weight
1	1988F1	F	1988	14	0.106
2	1988F1	F	1988	21	0.261
3	1988F1	F	1988	28	0.666
4	1988F1	F	1988	35	2.110
5	1988F1	F	1988	42	3.560
6	1988F1	F	1988	49	6.230

```
p <- ggplot(Soybean, aes(x = Time, y = weight)) +
  geom_point() + facet_wrap(~ Plot, ncol = 8) +
  labs(x = "Time (days after planting)",
       y = "Weight (average leaf weight per plant in grams)") +
  theme_minimal()
plot(p)
```

```
p <- ggplot(Soybean, aes(x = Time, y = weight)) +
  geom_point(size = 1) + facet_wrap(~ Variety) +
  geom_line(aes(group = Plot), linewidth = 0.1) +
  labs(x = "Time (days after planting)",
       y = "Weight (average leaf weight per plant in grams)") +
  theme_minimal()
plot(p)
```



Consider a logistic growth model which can be written as

$$E(W) = \frac{\theta_1}{1 + e^{-(t - \theta_2)/\theta_3}},$$

where θ_1 is the asymptote as $t \rightarrow \infty$, θ_2 is the time at which the expected weight is $\theta_1/2$, and θ_3 is inversely related to the steepness of the curve at θ_2 . We could assume that each parameter varies by plot, and is also affected by variety as follows.

```
m <- nlme(weight ~ theta1 / (1 + exp(-(Time - theta2) / theta3)), data = Soybean,
  fixed = theta1 + theta2 + theta3 ~ Variety,
  random = theta1 + theta2 + theta3 ~ 1 | Plot,
  start = c(20, 0, 60, 0, 10, 0),
  control = nlmeControl(msMaxIter = 1000))
summary(m)$tTable
```

	Value	Std.Error	DF	t-value	p-value
theta1.(Intercept)	16.9466	1.0308	359	16.4410	1.121e-45
theta1.VarietyP	4.5663	1.4630	359	3.1212	1.947e-03
theta2.(Intercept)	54.8758	1.0561	359	51.9624	4.233e-169
theta2.VarietyP	0.1828	1.4504	359	0.1260	8.998e-01
theta3.(Intercept)	8.2286	0.4748	359	17.3324	2.478e-49
theta3.VarietyP	0.3740	0.6345	359	0.5895	5.559e-01

In more complex models getting the inferences you want from a `nlme` object can be a bit tricky. Functions like `contrast` and `emmeans` will not work with a `nlme` object. But you can use the `lincon` function, although you need to tell it how to extract the parameter estimates from `nlme` (it needs to use the `fixef` function). Here we can get results like those returned by `summary`.

```
trtools::lincon(m, fest = fixef)
```

	estimate	se	lower	upper	tvalue	df	pvalue
theta1.(Intercept)	16.9466	1.0232	14.9412	18.952	16.5621	Inf	1.310e-61
theta1.VarietyP	4.5663	1.4523	1.7198	7.413	3.1441	Inf	1.666e-03
theta2.(Intercept)	54.8758	1.0483	52.8210	56.930	52.3450	Inf	0.000e+00
theta2.VarietyP	0.1828	1.4398	-2.6392	3.005	0.1269	Inf	8.990e-01
theta3.(Intercept)	8.2286	0.4713	7.3049	9.152	17.4600	Inf	2.888e-68
theta3.VarietyP	0.3740	0.6298	-0.8604	1.608	0.5939	Inf	5.526e-01

The estimate of mean θ_1 parameter for the F variety is given by `theta1.(Intercept)`. But the estimate of the mean θ_1 parameter for the P variety is the sum of the `theta1.(Intercept)` and `theta1.VarietyP` parameters. This can be obtained as follows.

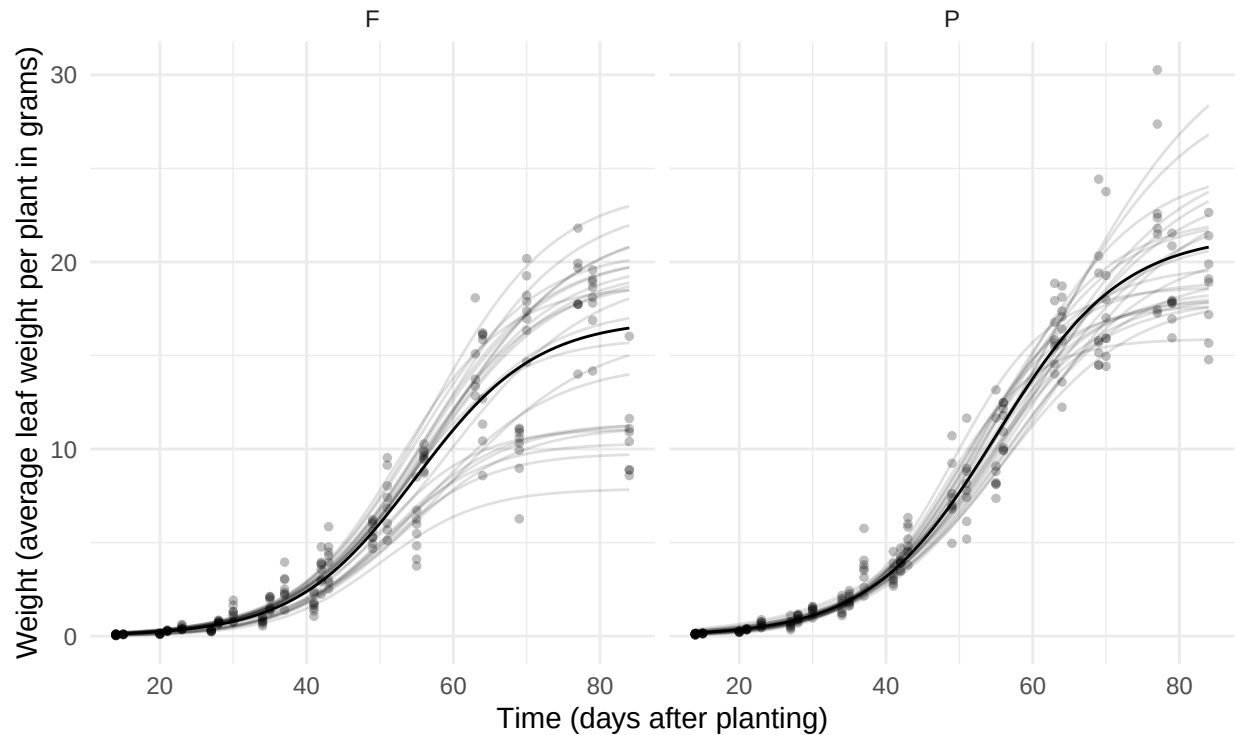
```
trtools::lincon(m, a = c(1,1,0,0,0,0), fest = fixef)
```

	estimate	se	lower	upper	tvalue	df	pvalue
(1,1,0,0,0,0),0	21.51	1.031	19.49	23.53	20.87	Inf	9.496e-97

Again we can plot this model as we did with the `Loblolly` data/model, although setting up the data frame is a little more complicated because plots and variety are not crossed.

```
library(dplyr)
library(tidyr)
d <- Soybean %>% dplyr::select(Plot, Variety) %>% unique() %>%
  group_by(Plot, Variety) %>% tidyr::expand(Time = seq(14, 84, length = 100))
d$yhat.ind <- predict(m, newdata = d, level = 1)
d$yhat.avg <- predict(m, newdata = d, level = 0)

p <- ggplot(Soybean, aes(x = Time, y = weight)) +
  geom_line(aes(y = yhat.ind, group = Plot), data = d, alpha = 0.125) +
  geom_line(aes(y = yhat.avg), data = d) +
  geom_point(size = 1, alpha = 0.25) + facet_wrap(~ Variety) +
  labs(x = "Time (days after planting)",
       y = "Weight (average leaf weight per plant in grams)") + theme_minimal()
plot(p)
```

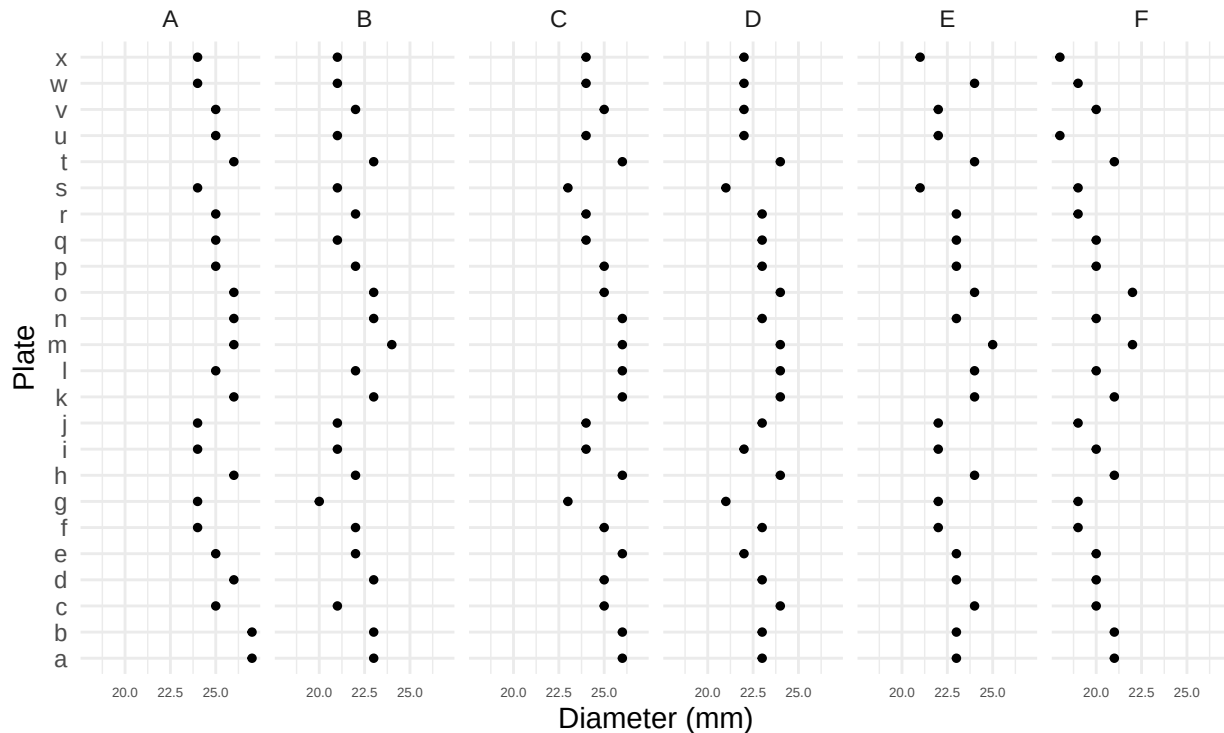


Crossed Random Effects

Crossed random effects might be specified when two (or more) factors modeled as having random effects are crossed (i.e., having a “factorial design” structure).

Example: Six samples of penicillin were tested using 24 plates. The response variable was the diameter of the zone of inhibition of the growth of a bacteria.

```
p <- ggplot(Penicillin, aes(x = plate, y = diameter)) +
  geom_point(size = 1) + facet_wrap(~ sample, ncol = 6) +
  coord_flip() + theme_minimal() +
  theme(axis.text.x = element_text(size = 5)) +
  labs(y = "Diameter (mm)", x = "Plate")
plot(p)
```



Let Y_{ij} denote the diameter of inhibition for the i -th sample ($i = 1, 2, \dots, 6$) and the j -th plate ($j = 1, 2, \dots, 26$). A model might be

$$E(Y_{ij}) = \beta_0 + \delta_i + \zeta_j \quad \text{or} \quad Y_{ij} = \beta_0 + \delta_i + \zeta_j + e_{ij}.$$

where δ_i and ζ_j are sample-specific and plate-specific effects, respectively. Here we will model both as random effects, each with an independent normal distribution. Note that we don't have any fixed effects.

```
m <- lmer(diameter ~ (1 | plate) + (1 | sample), data = Penicillin)
summary(m)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: diameter ~ (1 | plate) + (1 | sample)
Data: Penicillin
```

```
REML criterion at convergence: 330.9
```

```
Scaled residuals:
```

```
      Min       1Q   Median       3Q      Max
-2.0792 -0.6714  0.0629  0.5838  2.9796
```

```
Random effects:
```

```
Groups   Name             Variance Std.Dev.
plate    (Intercept)  0.717      0.847
sample   (Intercept)  3.731      1.932
Residual                    0.302      0.550
```

```
Number of obs: 144, groups: plate, 24; sample, 6
```

```
Fixed effects:
```

```
              Estimate Std. Error t value
(Intercept)   22.972      0.809    28.4
```

Example: Consider the following data from a study that examined mating success with multiple combinations of male and female salamanders.

```
library(hglm.data)
data(salamander)
head(salamander, 12)
```

	Season	Experiment	TypeM	TypeF	Cross	Male	Female	Mate
1	Summer	1	R	R	RR	1	1	1
2	Summer	1	W	R	RW	14	1	1
3	Summer	1	R	R	RR	5	1	1
4	Summer	1	W	R	RW	11	1	0
5	Summer	1	R	R	RR	4	1	1
6	Summer	1	W	R	RW	15	1	1
7	Summer	1	R	R	RR	5	2	1
8	Summer	1	W	R	RW	15	2	1
9	Summer	1	R	R	RR	3	2	1
10	Summer	1	W	R	RW	13	2	1
11	Summer	1	R	R	RR	1	2	1
12	Summer	1	W	R	RW	12	2	1

The question here is how the combination female and male salamanders in terms of population (W = “White Side”, R = “Rough Butt”) affects mating success, while accounting for individual differences in the salamanders themselves.

```
salamander$Cross <- relevel(salamander$Cross, ref = "WR")
m <- glmer(Mate ~ Cross + (1 | Male) + (1 | Female),
  family = binomial, data = salamander)
summary(m)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) ['glmerMod']
Family: binomial ( logit )
Formula: Mate ~ Cross + (1 | Male) + (1 | Female)
Data: salamander
```

AIC	BIC	logLik	deviance	df.resid
430.6	453.9	-209.3	418.6	354

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.051	-0.616	0.271	0.597	2.551

Random effects:

Groups	Name	Variance	Std.Dev.
Male	(Intercept)	1.04	1.02
Female	(Intercept)	1.17	1.08

Number of obs: 360, groups: Male, 60; Female, 60

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.896	0.446	-4.25	2.1e-05 ***
CrossRR	2.904	0.561	5.18	2.2e-07 ***
CrossRW	2.202	0.588	3.75	0.00018 ***
CrossWW	2.886	0.549	5.26	1.4e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
Correlation of Fixed Effects:
      (Intr) CrssRR CrssRW
CrossRR -0.716
CrossRW -0.771  0.678
CrossWW -0.709  0.521  0.661
```

Nested Random Effects

Nested factors occur when they form a *hierarchical* structure. For example, in the **Sitka** data the levels of **tree** are nested within levels of **treat** (i.e., ozone or control), and in the **Soybean** data the levels of **Plot** are nested within levels of **Variety**. Nested random effects when the levels of one factor, modeled as a random effect, are nested within the levels of another factor that is also modeled as a random effect.

Example: The **Pastes** data frame from the **lme4** package is from a study of the strength of chemical pastes. Paste was delivered in a cask, and there were three casks per batch, and 10 batches. Two tests were run per cask.

```
head(Pastes, 12)
```

	strength	batch	cask	sample
1	62.8	A	a	A:a
2	62.6	A	a	A:a
3	60.1	A	b	A:b
4	62.3	A	b	A:b
5	62.7	A	c	A:c
6	63.1	A	c	A:c
7	60.0	B	a	B:a
8	61.4	B	a	B:a
9	57.5	B	b	B:b
10	56.9	B	b	B:b
11	61.1	B	c	B:c
12	58.9	B	c	B:c

Note that levels of **cask** are not the same between batches — i.e., cask **a** in batch **A** is *not* the same as cask **a** in batch **B**, for example. The **sample** variable was created to identify a particular **cask**. We could model these data as

$$E(Y_{ijk}) = \beta_0 + \delta_i + \zeta_{ij},$$

where Y_{ijk} is k -th ($k = 1, 2$) test a paste from the j -th ($j = 1, 2, 3$) cask from the i -th batch ($i = 1, 2, \dots, 10$). So here δ_i is the effect of the i -th batch, and ζ_{ij} is the effect of the j -th cask from the i -th batch.

```
m <- lmer(strength ~ (1|batch) + (1|cask:batch), data = Pastes)
summary(m)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: strength ~ (1 | batch) + (1 | cask:batch)
Data: Pastes
```

```
REML criterion at convergence: 247
```

```
Scaled residuals:
```

	Min	1Q	Median	3Q	Max
	-1.4798	-0.5156	0.0095	0.4720	1.3897

```
Random effects:
```

Groups	Name	Variance	Std.Dev.

```

cask:batch (Intercept) 8.434    2.904
batch      (Intercept) 1.657    1.287
Residual                   0.678    0.823
Number of obs: 60, groups:  cask:batch, 30; batch, 10

```

Fixed effects:

```

              Estimate Std. Error t value
(Intercept)   60.053      0.677    88.7

```

Note that you could use `sample` in place of `cask:batch`.

Example: Consider the following data from an experiment on the treatment of arthritis.

```

myhips <- faraway::hips %>% pivot_longer(cols = c(fbef,faft,rbef,raft),
  names_to = "obs", values_to = "angle") %>%
  mutate(time = rep(c("before","after"), n()/2)) %>%
  mutate(variable = rep(c("flexion","flexion","rotation","rotation"), n()/4)) %>%
  mutate(time = factor(time, levels = c("before","after")))
head(myhips,10)

```

A tibble: 10 x 7

```

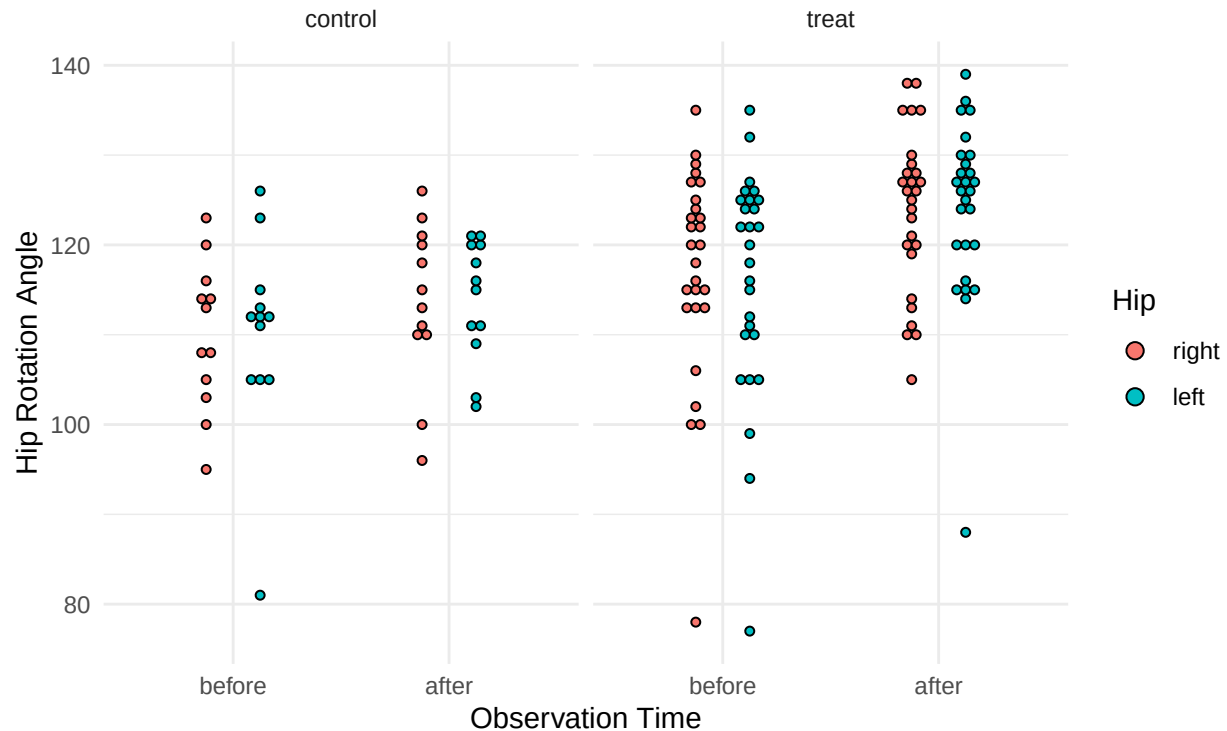
  grp  side person obs  angle time  variable
<fct> <fct> <fct> <chr> <dbl> <fct> <chr>
1 treat right 1     fbef  125 before flexion
2 treat right 1     faft  126 after  flexion
3 treat right 1     rbef   25 before rotation
4 treat right 1     raft   36 after  rotation
5 treat left  1     fbef  120 before flexion
6 treat left  1     faft  127 after  flexion
7 treat left  1     rbef   35 before rotation
8 treat left  1     raft   37 after  rotation
9 treat right 2     fbef  135 before flexion
10 treat right 2     faft  135 after  flexion

```

```

p <- ggplot(subset(myhips, variable == "flexion"), aes(x = time, y = angle, fill = side)) +
  theme_minimal() + geom_dotplot(binaxis = "y", stackdir = "center", binwidth = 1,
  position = position_dodge(width = 0.5)) + facet_wrap(~ grp) +
  labs(x = "Observation Time", y = "Hip Rotation Angle", fill = "Hip")
plot(p)

```

Here for each of two response variables (flexion and rotation) we have two observations (before and after) for each hip (side) for each person. Here we specify a random effect for each person and a random effect for each hip within each person. Here we will consider the rotation response variable. Note that I am assuming that there is not, on average, an effect of left versus right side.

```
m <- lmer(angle ~ time * grp + (1|person) + (1|person:side),
  subset = variable == "rotation", data = myhips)
summary(m)
```

Linear mixed model fit by REML ['lmerMod']

Formula: angle ~ time * grp + (1 | person) + (1 | person:side)

Data: myhips

Subset: variable == "rotation"

REML criterion at convergence: 1033

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.0275	-0.5006	0.0254	0.4548	1.8289

Random effects:

Groups	Name	Variance	Std.Dev.
person:side	(Intercept)	33.1	5.76
person	(Intercept)	27.6	5.25
Residual		18.0	4.24

Number of obs: 156, groups: person:side, 78; person, 39

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	25.000	2.105	11.88

timeafter	0.958	1.224	0.78
grptreat	-0.222	2.529	-0.09
timeafter:grptreat	5.634	1.471	3.83

Correlation of Fixed Effects:

	(Intr)	timftr	grptrt
timeafter	-0.291		
grptreat	-0.832	0.242	
tmftr:grp	0.242	-0.832	-0.291

What is the estimated change in expected rotation from before to after treatment in each group?

```
trtools::contrast(m,
  a = list(time = "after", grp = c("control","treat")),
  b = list(time = "before", grp = c("control","treat")),
  cnames = c("control","treat"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
control	0.9583	1.2238	-1.440	3.357	0.783	Inf	4.336e-01
treat	6.5926	0.8159	4.993	8.192	8.080	Inf	6.468e-16

The `icc_specs` function from the **specr** package can be used to produce estimates concerning the “variance components” (i.e., the variance due to person, side, and error).

```
specr::icc_specs(m)
```

	grp	vcov	icc	percent
1	person:side	33.14	0.4210	42.10
2	person	27.59	0.3506	35.06
3	Residual	17.97	0.2284	22.84